Bioresource Technology 320 (2021) 124332

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Effect of different aerobic hydrolysis time on the anaerobic digestion characteristics and energy consumption analysis

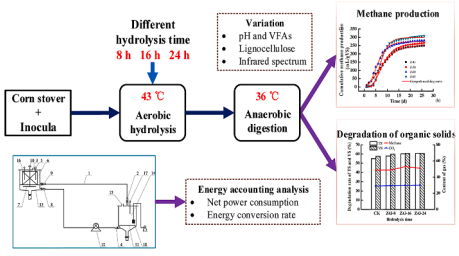
Jingbo Qu [a,b,c], Yong Sun [a,c], Mukesh Kumar Awasthi [d], Yuyingnan Liu [a,c], Xinrui Xu [a,c], Xianghui Meng [a,c], Hongqiong Zhang [a,c,e,*]

[a] College of Engineering, Northeast Agriculture University, Harbin 150030, PR China
[b] Guangdong Provincial Key Laboratory of New and Renewable Energy Research and Development, Guangzhou 510640, PR China
[c] Key Laboratory of Agricultural Renewable Resources Utilization Technology and Equipment in Cold Areas of Heilongjiang Province, Harbin 150030, PR China
[d] College of Natural Resources and Environment, Northwest A&F University, Taicheng Road 3#, Yangling, Shaanxi 712100, PR China
[e] Key Laboratory of New Materials and Facilities for Rural Renewable Energy of Ministry of Agriculture and Rural Affairs, Henan Agricultural University, Zhengzhou 450002, PR China

HIGHLIGHTS

- The accumulation rate of VFAs began to slow down after 16 h of hydrolysis.
- The degradation rate of lignocellulose was obviously increased after hydrolysis.
- 16 h of hydrolysis had the best cumulative methane production with 268.75 ml/g VS.
- 16 h of hydrolysis can obtain best energy conversion efficiency, especially in July.
- Problem of over degradation of digestion raw materials was solved by this strategy.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Corn stover
Aerobic hydrolysis
Anaerobic digestion
Methane production
Energy consumption

ABSTRACT

Aerobic hydrolysis of stover before anaerobic digestion is beneficial to improve the biodegradability of corn stover. Aerobic hydrolysis of corn stover at 43 °C was conducted to investigate the effects of hydrolysis time (0 h, 8 h, 16 h, and 24 h) on the degradation of lignocellulose from corn stover and material conversion. Further anaerobic digestion and energy consumption analysis with the digestion temperature of 36 °C were carried out. The accumulation rate of volatile fatty acids began to slow down after 16 h of hydrolysis, and the concentration of acetic acid reached 221.85 mmol/L at 24 h of hydrolysis. The degradation rate of lignocellulose was obviously increased after hydrolysis. When the hydrolysis time was 16 h, it reached the maximum cumulative methane production with 268.75 ml/g VS. In terms of biogas production and energy conversion efficiency, it is more appropriate to choose 16 h as hydrolysis time in biogas engineering.

* Corresponding author at: College of Engineering, Northeast Agricultural University, Harbin 150030, PR China.
E-mail address: zhhqiong@163.com (H. Zhang).

https://doi.org/10.1016/j.biortech.2020.124332
Received 7 September 2020; Received in revised form 23 October 2020; Accepted 24 October 2020
Available online 29 October 2020
0960-8524/© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

As a good fermentation material, crop stover has low sulfur content and high methane production, which is an important raw material source for the large-scale development of biogas engineering (Sun et al., 2020b; Wang et al., 2019). In the production of biogas from corn stover (CS), due to the complex cross-linking structure of lignocellulose and the high crystallinity of cellulose molecules, it is difficult for the stover to be directly and efficiently degraded by microorganisms (Sun et al., 2018; Wainaina et al., 2020a). Therefore, improving the efficiency of hydrolysis and acidification of stover is the key factor to improve the biogas fermentation efficiency. In order to improve methane production in the fermentation process, it is necessary to pretreat CS properly. The pretreatment methods of CS can be roughly divided into physical, physical-chemical, chemical, and biological pretreatment, which have their own advantages and disadvantages (Wainaina et al., 2019; Zhang et al., 2020). Most of the current researches are still in the small experiment stage with complicated process operation, which is difficult to apply in large-scale industrial production (Bremond et al., 2018; Yu et al., 2019; Wainaina et al., 2020b). Therefore, finding a pretreatment method with wide adaptability, strong operability, and low cost is still the focus and difficulty of research.

Two-stage anaerobic digestion (AD) is a common type of anaerobic biological treatment process (Srisowmeya et al., 2020), which includes two parts: aerobic hydrolysis (AH) fermentation phase and anaerobic methanogenic fermentation phase. The AH stage mainly uses the action of aerobic microorganism and oxygen molecular to increase the degradation rate of organic matter in the feed solution. In AD stage, methanogens convert the small molecules in aerobic phase into methane. The microorganisms in the above-mentioned two stages are quite different, and each aerobic microorganism has its own specific object and suitable conditions. The reproductive capacity of aerobic microorganisms is much higher than that of anaerobic microorganisms, and lignin can be further degraded by microorganisms through the variation of side chain under the action of oxygen molecular (Li et al., 2018; Yang et al., 2019b).

AH technology is to decompose insoluble macromolecular organic matter into small molecular organic matter under the action of extracellular enzymes, which provides substrate for subsequent fermentation bacteria (Du and Li, 2017; Kavitha et al., 2016; Schiltz et al., 2020). Lim et al. indicated that aerobic aeration in the AD system could improve the efficiency of hydrolysis and acidification, enhance the conversion of short-chain fatty acids (SCFAs) to acetic acid, which was due to the enhancement of activities of acidogenic and hydrolytic bacteria and the slow degradation of compounds under micro-oxygen conditions (Lim and Wang, 2013; Wainaina et al., 2019). The suitable oxygen supply rate is the key factor to ensure the good effect of AD system (Park et al., 2019). By controlling the intensity of aerobic aeration, the influence of excess oxygen on strictly anaerobic environment can be avoided. Moreover, the two-stage AD process can promote the stability of the system. Through the control of the acidogenic phase, balanced nutrition can be provided for the methanogenic phase, which avoids overload or inhibition of methanogenic phase (Shi et al., 2020).

In the hydrolysis stage, it is the key problem to solve the high efficiency AD of CS to destroy the lignin structure and make the extracellular enzymes in contact with cellulose and hemicellulose (Awasthi et al., 2020; Wu et al., 2018). Ma et al. indicated that the degradation rates of lignin and hemicellulose could both reach more than 9% and 28% after 24 h of hydrolysis, respectively (Ma et al., 2017; Wainaina et al., 2019). However, the changes of microbial species in the process of hydrolysis need to be further analyzed and discussed. Li et al. found that the average methane yield of the reactor after AH was 0.13 m^3/kg (Volatile solids (VS)), which was significantly higher than that of the control group (Li and Pagilla, 2017). Similarly, De la Rubia et al. showed that in aerobic pretreatment experiments with different microorganisms, the biogas production of olive oil plant wastewater treated with aspergillus was 35% higher than that of the control group (De la Rubia et al., 2018; Wainaina et al., 2019). From these studies, it can be seen that AH treatment has a positive effect on AD. In the current research, this treatment method is more applied to the treatment of wastewater pollutants, and the research on its application in AD of CS is still less. Moreover, the effect of AH time on AD characteristics and energy consumption analysis of digestion system have not been explored, and AD efficiency is seriously restricted in cold regions of China by the long fermentation time of CS and the over degraded organic matter.

In this study, the characteristics of AH of CS were explored, and the degradation of lignocellulose, the change of solid organic matter content, and material conversion were investigated. Then, the effects of different AH time on the AD performance of CS were analyzed, and the modified Gompertz model was used to predict the methane production potential. Moreover, based on the 100 m^3 biogas engineering, the proportion of net power consumption and energy conversion rate of AD system were studied, and the relationship between productivity and energy consumption in different AH time was analyzed. This study can provide important theoretical and practical value to the rapid degradation and efficient utilization of CS and solve the problem of raw material treatment in biogas engineering.

2. Materials and methods

2.1. Feed stocks and inoculum

The CS was obtained in the Xiangyang Experimental Base of Northeast Agricultural University of Heilongjiang Province, China. The naturally dried CS was crushed by the hammer mill (Sida, China, 5.5 kW) with the screen perforation of 5 mm. Inocula were obtained from the anaerobic digester of cattle manure at 35 ± 2 °C, and the size and hydraulic retention time (HRT) of digester were 5400 m^3 and 30 d. Moreover, the detailed properties of CS and inocula were presented in Table 1.

2.2. Experimental design

The experiment was divided into two stages: AH and AD. According to the results obtained by the research group, 43 °C was selected as AH temperature in this experiment (Ma et al., 2017). A total of 21 jars with an effective volume of 1750 ml were prepared as the reactor in this experiment, and nine jars labeled from 1 to 9 were used in the AH stage to detect the changes of material composition in this process. Another 12 jars labeled from 10 to 21 were used in the AD stage, four treatments with different hydrolysis time were CK (single AD without AH, jar 10 to 12), hydrolysis for 8 h (Z43-8, jar 13 to 15), hydrolysis for 16 h (Z43-16, jar 16 to 18), and hydrolysis for 24 h (Z43-24, jar 19 to 21), respectively, and three parallel replicates were performed.

Corn stover was mechanically crushed, and the TS and VS of the crushed stover and biogas slurry, and pH of the biogas slurry were measured. At the same time, part of the crushed CS and biogas slurry were stored in sealed bags and centrifugal tubes respectively for component detection. 80 g CS and 900 g inoculum (TS of reactants was 8%) were loaded into each of above 21 jars, and one aeration device was put into each reactor except jar 10, 11, and 12. The aeration device adopted an electromagnetic vibration air pump, with 198 mm in length, 145 mm in width, and 96 mm in height, and the aeration rate was set to 2.5 L/ min. Each reactor was put into a 43 °C water bath for heating, and the aerator was turn on to start aeration. During the process of the reaction, each reactor was stirred manually every 4 h, and samples were taken from jar 1 to 9 for the AH stage to be measured according to time changes. The sample was used for the detection and analysis of the components in the hydrolysis process, which included pH, TS, VS, lignocellulose, volatile fatty acids (volatile fatty acid), and infrared spectrum.

The time of aerobic hydrolysis was controlled as a variable. The

Table 1
Characteristic of corn stover and inoculum.

Material	Total Solids (%)	Volatile solids (%)	Carbon (% of TS)	Nitrogen (% of VS)	Cellulose (%)	Hemicellulose (%)	Lignin (%)	pH
Corn stover	92.48 ± 0.89	86.24 ± 1.02	41.88 ± 0.47	0.57 ± 0.01	37.46 ± 0.30	20.26 ± 0.12	18.18 ± 0.05	NA
Inoculum	1.51 ± 0.12	0.87 ± 0.08	29.13 ± 0.03	1.54 ± 0.01	NA	NA	NA	7.41 ± 0.02

NA – not analyse.

aeration device in the jar 13 to 21 was taken out after the AH for 8 h, 16 h, and 24 h, respectively, and then AD experiment was carried out in jar 13 to 21. According to the results obtained by the research group, 36 °C was selected as AD temperature in this experiment (Ma et al., 2017; Yang et al., 2019a). The inoculation material was selected from the completely domesticated biogas slurry. The pH of the hydrolysate after AH was adjusted with the same amount of biogas slurry (the appropriate range was 6.5–7). Nitrogen was introduced to remove the oxygen in the hydrolysate, and the bottle mouth of the anaerobic reactor was sealed with rubber plug and glass glue. The reactor was connected with a 2 L air collecting bag and placed in 36 °C water bath for AD experiment. During the experiment, the air collecting bag was removed regularly every day to determine the gas composition and volume. Moreover, the fermentation broth was manually shaken twice a day to make the raw materials fully contact with the microorganisms in the inoculum.

2.3. Analytical methods

The analysis of TS, VS, and pH was used the APHA Standard Method (Miner, 2006). Lignocellulose was determined through the Van Soest process by the fiber instrument (Ankom 200i, Anke Borui, Beijing, China), and correlative analysis procedure were as follows: analyzing the content of neutral detergent fiber, the content of acid detergent fiber, and the content of acid detergent lignin (Van Soest, 1967). Carbon (C) content and nitrogen (N) content were analyzed by the instrument of elemental analysis (EA 3000, LEEMAN, Beijing, China). Biogas and VFA were analyzed using the gas chromatography (GC-6890N, Agilent, USA). Biogas was determined through the thermal conductivity detector. VFA was determined through the flame ionization detector and capillary column. Each treatment was repeated three times. The functional groups were investigated using infrared spectrometer (TENSOR 27, Bruker Co., Ltd., Germany) with the temperature of 20 °C and the scanning range from 400 cm^{-1} to 4000^{-1}.

2.4. Model methods

The first order model and the modified Gompertz model were applied to accurately analyze and predict the methane production capacity of AD by fitting the experimental data of methane production (Yang et al., 2019a). The modified Gompertz model is shown as the Eq. (1).

$$P(t) = P_\infty \times exp\left\{ -exp\left[\frac{R_m \times e}{P_\infty}(\lambda - t) + 1\right]\right\} \quad (1)$$

in which $P(t)$ expresses the cumulative methane production at time t (d) (mL/gvolatilesolid(VS)), P_∞ expresses the maximum methane production potential (mL/gVS), R_m represents for the rate of maximum methane production (mL/gVSÂ·d), λ expresses the time of lag period (d), t expresses the time over the period of AD, and e is a constant with the value of 2.718. And the First-order model is shown as the Eq. (2) (Yang et al., 2019a).

$$r = \mathrm{In}\left(\frac{C_{S0}}{C_S}\right) \times C_S/t \quad (2)$$

in which r expressses the rate of methane production at time t (mL/gVSÂ·d), C_{S0} expresses the maximum methane production (mL/gVS), C_S represents for the difference between methane production at time t and that of maximum value (mL/gVS), and t expresses the AD time (d).

2.5. Analysis of energy input and output

Static energy consumption analysis method was used to study the energy consumption of biogas engineering, and the influence of aerobic pretreatment on energy consumption of AD system was investigated. The main accounting of energy consumption included raw materials crushing, feed solution warming, and heating energy consumption in hydrolysis and AD process to maintain normal operation of the system, and the energy consumption of mixing, material transportation and solid-liquid separation was not considered. The structure of biogas production system of two-phase AD in this study is shown in Fig. 1, and both the AH reactor and the AD reactor were 100 m^3. The substrate was placed in the AH device in this system. The hydrolysis temperature was controlled to 43 °C by heating and heat preservation device. The AH device was equipped with self-priming centrifugal aeration device, with 1050 mm in length, 840 mm in width, and 1100 mm in height, and the aeration rate was set to 4.95 m^3/ min. The organic matter in the substrate produced organic digestion liquid (acetic acid, propionic acid or butyric acid and other organic acids) under the action of hydrolysis bacteria and acid-producing bacteria. At the same time, the gas was released through the outlet of the AH device, and the organic digestion liquid was filtered by the perforated baffle and then pumped into the AD device with the circulating liquid. The heating device and the outer wall insulation layer in the AD device controlled the fermentation temperature at 36 °C, and the organic digestion liquid was converted into methane by methanogens in the AD device. For the biogas project, according to the law of energy conservation, the heat loss of biogas project should be equal to the energy input from outside. The heat loss of biogas project mainly includes the heat required for the heating of new fermentation raw materials and the heat consumption of various

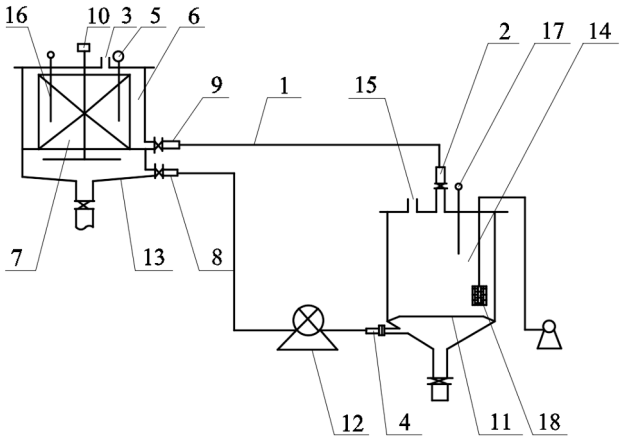


Fig. 1. Reactor structure; Note: 1 is the circulating tube, 2 and 8 are feed solution inlets, 3 and 15 are gas outlets, 4 and 9 are feed solution outlets, 5 is the temperature control probe, 6 is the effluent weir, 7 is the fermentation material, 10 is the mixing device, 11 is the perforated baffle, 12 is the lift pump, 13 is the integrated anaerobic digestion device, 14 is the integrated aerobic hydrolysis device, 16 and 17 are heating devices, and 18 is the aeration device.

reactors. The heat loss analysis methods of biogas project are as follows.

2.5.1. Heating consumption of fermentation raw material

The temperature of fermentation raw material is generally lower than that required for AD. According to the process of AD system, the fermentation raw materials are first mixed with part of the biogas slurry to be heated up before being input into the hydrolysis reactor, and then heated to the hydrolysis temperature (Duan et al., 2019). According to the law of conservation of energy, the calculation of temperature rise after the mixture of fermentation material and biogas slurry was demonstrated as Eq. (3).

$$t_H = \frac{m_y c_y t_y + m_L c_L t_L}{m_y c_y + m_L c_L} \tag{3}$$

in which m_y expresses the amount of fermenting raw materials (kg), c_y expresses the specific heat capacity of fermenting raw materials [1.62 kJ/(kg·K)], t_y expresses the initial temperature of fermenting raw materials (15 °C), m_L expresses the return flow of biogas slurry (kg), c_L expressed the specific heat capacity of biogas slurry [4.20 kJ/(kg·K)], t_y expresses the temperature of biogas slurry (30 °C), and t_H expresses the temperature of the mixed feed solution (℃).

The calculation of heat required for mixed feed solution was shown as Eq. (4).

$$Q_H = (m_y + m_L) c_f (t_S - t_H) \tag{4}$$

in which Q_H expresses the heat required for raising the temperature of the mixed feed solution (kJ), c_f expresses the specific heat capacity of the mixed feed solution [4.20 kJ/(kg·K)], and t_S is hydrolysis temperature (43 °C).

2.5.2. Heating consumption of AD reactor

Under the condition of constant temperature of feed solution in AD reactor, the heat balance equation of the reactor was shown as Eq. (5).

$$Q_{hf} + Q_n + Q_j = Q_f + Q_g + Q_W + Q_c \tag{5}$$

in which Q_{hf} expresses the heat output of the heating system (kJ), Q_n expresses the biochemical heat from the fermentation (kJ), Q_j expresses the energy brought into the reactor by fermenting feed solution (kJ), Q_f expresses the heat consumption of the reactor enclosure structure (KJ), Q_g expresses the sensible heat loss from biogas exhausting reactor (KJ), Q_W expresses the heat loss of water evaporation in the reactor (KJ), and Q_c expresses the energy brought out of the reactor by fermenting feed solution (kJ).

In the actual biogas engineering, in order to keep the amount of feed solution in the reactor constant, the volume flow of feed solution entering and leaving the reactor is equal (Bayrakdar et al., 2016). Due to the temperature of the feed solution entering and leaving the reactor is the same, and the density and specific heat capacity of the feed solution before and after the AD reaction have little change, it can be concluded that $Q_j \approx Q_c$. Considering that the activity of anaerobic microorganism is not strong enough and the heat released is very little, Q_n can be ignored. Therefore, Eq. (5) can be simplified to Eq. (6).

$$Q_{hf} = Q_f + Q_g + Q_W \tag{6}$$

To calculate the heat consumption of AD reactor, the following simplifications and assumptions are made. The reactor is a cylindrical structure above ground; the influence of radiation heat transfer between the reactor surface, the sun, and the surrounding environment is ignored; the temperature of feed solution in the reactor is uniform and constant; the temperature of the wall in contact with the feed solution in the reactor is 36 °C; the temperature of the untouched wall is about 3–8 °C lower than that of the feed solution (Peng et al., 2020), which is taken as 30 °C in this study. It is assumed that the AD reactor is a constant temperature heat source and dissipates heat through the insulation layer. The heat consumption of the enclosure structure mainly includes three parts: heat consumption at the top of the reactor, heat consumption on the side wall of the reactor, and heat consumption on the ground at the bottom of the reactor. The heat consumption of each part of the reactor in heat transfer was demonstrated as Eq. (7).

$$Q_{f1} = 86.4KA(t_{n1} - t_w)\tau \tag{7}$$

in which Q_{f1} expresses monthly heat consumption of heat transfer on top of reactor (kJ), K expresses the heat transfer coefficient of reactor (W/(m^2·K)), A is the reactor area (m^2), t_{n1} expresses the reactor temperature (℃), t_w is average monthly outdoor temperature (℃), τ is total days per month (d), and 86.4 expresses the unit conversion coefficient. The calculation of heat loss of water vapor was presented as Eq. (8).

$$Q_W = 0.001 W_\omega \left[H_v + c_p (t_f - t_w)\right]\tau \tag{8}$$

in which Q_W expresses monthly heat loss of water evaporation (kJ), W_ω expresses the mass flow rate of water vapor in the biogas flow out of the reactor (kg/d), H_v expresses the latent heat of vaporization at anaerobic fermentation temperature (2417.8 kJ/kg), c_p expresses the specific heat capacity at constant pressure of water vapor at anaerobic fermentation temperature [1.867 kJ/(kg·K)], and t_f is the temperature of anaerobic fermentation (℃). The heat consumption of the side wall of the reactor was presented as Eq. (9) (Li et al., 2019).

$$Q_{f2} = 86.4K_2A_2(t_{n2} - t_w)\tau \tag{9}$$

in which Q_{f2} expresses monthly heat consumption of heat transfer on the side wall of reactor (kJ), K_2 expresses the heat transfer coefficient of the side wall of reactor (W/(m^2·K)), A_2 is area of the side wall of reactor (m^2), t_{n2} expresses the temperature of the side wall of reactor (℃), and other symbols have the same meaning as before. The heat consumption of the bottom of reactor was presented as Eq. (10) (Li et al., 2019).

$$Q_{f3} = 86.4[2\lambda_t(t_{n3} - t_w)][A_0 \ln\frac{A_0}{A_0 - r_f} - r_f]\tau \tag{10}$$

in which Q_{f3} expresses monthly heat consumption of heat transfer on the bottom of reactor (kJ), t_{n3} expresses the temperature of the bottom of reactor (℃), A_0 expresses the radius of the bottom of reactor (m), r_f expresses the internal radius of the bottom of reactor (m), and other symbols have the same meaning as before.

The heat loss of water evaporation was presented as Eq. (11).

$$Q_W = 0.001 W_\omega \left[H_V + C_p (t_f - t_w)\right]\tau \tag{11}$$

in which Q_W expresses monthly heat loss due to water evaporation (kJ), W_ω expresses the mass flow rate of water vapor in biogas flow out of reactor (kg/d), H_V expresses the latent heat of water vaporization at AD temperature [2417.8 (kJ/kg)], C_p expresses the specific heat capacity at constant pressure of water vapor at AD temperature [1.867 kJ/(kg·K)], t_f expresses the AD temperature (36 °C), and other symbols have the same meaning as before.

2.5.3. Heating consumption of hydrolysis reactor

Under the condition of constant temperature of feed solution in the hydrolysis reactor, the heat balance of the reactor was shown as Eq. (12).

$$Q_{hs} = Q_s + Q_q \tag{12}$$

in which Q_{hs} expresses the heat output of the heating system (kJ), Q_s expresses the heat consumption of the reactor enclosure structure (KJ), and Q_q expresses the heat consumption in aeration process (KJ).

The heat consumption in aeration process was calculated as Eq. (13).

$$Q_q = V\rho c(t_s - t_j)\tau \tag{13}$$

in which V is the aeration rate of aeration process (30 m^3/h), ρ is the air

density (1.293 kg/m^3), c is the specific heat capacity of air [1.004 kJ/(kg·K)], t_s is the hydrolysis temperature (43 °C), t_j is the intake temperature (15 °C), and τ is the aeration time (8 h, 16 h, and 24 h).

In the multi index evaluation system, due to the different property of the indicators, there are different dimensions and orders of magnitude. In the case of large differences in the level of indicators, when the original samples are directly used for comprehensive analysis, the role of indicators with higher values is often highlighted, while the role of indicators with lower values is ignored. However, some sensitive indicators are important indicators in the evaluation system, although their values are relatively low. In order to put all indexes in the evaluation system at the same order of magnitude for comprehensive evaluation, this study introduced numerical standardization. The standardized equation was calculated as Eq. (14).

$$S' = \frac{S - Min}{Max - Min} \tag{14}$$

where S is the original index, S′ is the normalization value, and Max and Min ae the maximum and minimum values of the original sample, respectively.

2.6. Data processing

Three parallel replicates were performed in this study. Excel 2007 was used for data collation, and Origin 8.5 software was employed for drawing. SPSS 22.0 software was used for one-way ANOVA to test the significance of the data and standardization of raw data.

3. Results and discussion

3.1. Analysis of parameters in aerobic hydrolysis

After aerobic warming to 43 °C, the CS was further AH, and the variation of the VFA and pH in the 24 h hydrolysis were shown in Fig. 2 (a). In the first 4 h, pH showed an upward trend, which was caused by the buffer material in the system maintaining the acid-base balance of the reactor, so that pH did not decrease with the rapid formation of organic acids (Zou et al., 2018). The optimum pH ranges of methanogens and hydrolytic acidification bacteria were not the same, and acidogenic bacteria could react at a lower pH. In this experiment, the pH value decreased rapidly from 6.92 to 5.86 within 4–8 h. Due to the rapid accumulation of VFA, the pH in the reactor began to decrease significantly, and an environment suitable for the survival of acidogenic bacteria was gradually formed (Ma et al., 2017; Wainaina et al., 2020a). At the late stage of AH (after 20 h), pH showed a slight upward trend. The reason is that in the process of AH, in addition to sugars consumed by acidogenic bacteria to produce organic acids, proteins and other substances are also used by ammonifying bacteria, and gradually converted to NH_4^+-N. The accumulation of ammonia substances made pH slightly higher in the later reaction period.

Due to the high initial moisture, CS was soaked, dissolved or precipitated easily degradable part of matrix after AH. The cumulative content of acetic acid was the highest within 24 h. The cumulative content of ethanol increased at first and then decreased. The ethanol content rapidly increased to 93.18 mmol/L in the first 12 h before hydrolysis, and then decreased rapidly within 12 h to 24 h. The cumulative content of butyric acid increased rapidly from 10.31 mmol/L to 84.31 mmol/L during 12 h to 24 h. The decreasing time of ethanol cumulative content and the increasing time of butyric acid cumulative content both started after 12 h, which was consistent with the change time of cumulative content of acetic acid. This is because during the hydrolysis process, acetic acid was not only produced by the extracellular enzymes of acidogenic bacteria, but also by organic acids such as ethanol, propionic acid and butyric acid. Wang indicated that butyric acid was gradually converted to acetic acid, and the concentration of acetic acid went up (Wang et al. 2020c). After 8 h, with the continuous utilization of methanogens, the acetic acid concentration decreased, which was inconsistent with the results of this study. In this study, after 12 h, butyric acid content began to accumulate gradually, and ethanol content gradually decreased, which indicated that after hydrolysis to 12 h the production of acetic acid began to change from consuming butyric acid to ethanol. Organic acids are gradually metabolized by acetogenic bacteria during the reaction, which is also the reason why the concentration of acetic acid was much higher than that of other organic acids during AH (Long et al., 2019; Wainaina et al., 2020b).

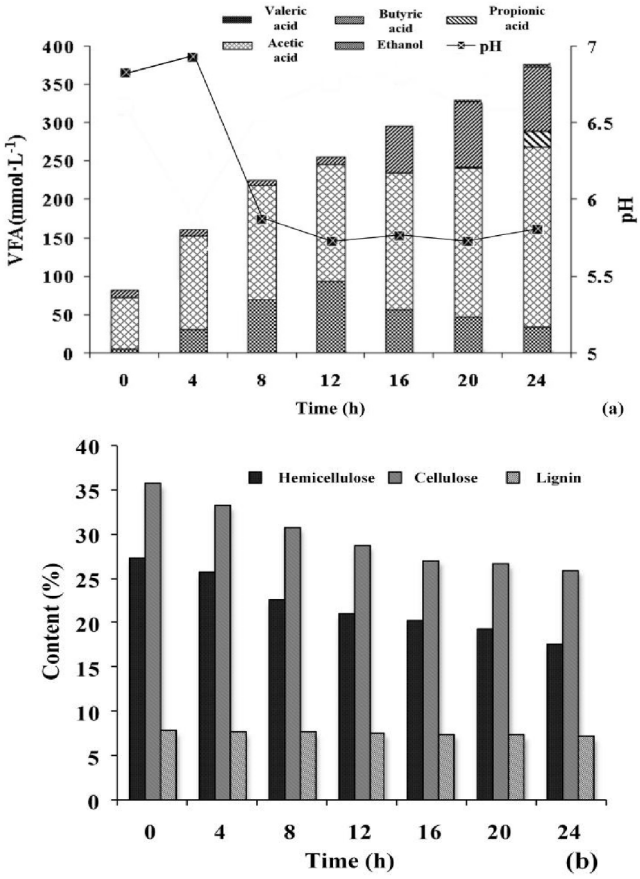


Fig. 2. Variation of parameters with aerobic hydrolysis time; (a) pH and VFAs, and (b) lignocellulose degradation rate.

The variation of lignocellulose degradation rate with AH time was shown in Fig. 2(b). After 4 h hydrolysis, the degradation rates of hemicellulose, cellulose, and lignin were 7.02%, 4.18%, and 3.33%, respectively. The warming energy of raw material came from the aerobic bacteria consuming the organic matter of CS and converting it into energy. After 24 h hydrolysis, the degradation rates of hemicellulose, cellulose, and lignin were 35.73%, 27.38%, and 8.06%, respectively. As the structure of CS was destroyed to some extent after AH, lignocellulose was consumed more. The variation of chemical composition and physical structure of CS can indirectly reflect the effect of AH (Sun et al., 2018, 2020a). In the early stage of the reaction, the aerobic microorganism made full use of oxygen. After 16 h, the product inhibition began to form gradually, which made the metabolic activity of microorganism begin to slow down and the consumption of oxygen began to weaken. At this time, oxygen molecules began to combine with lignin gradually, which showed the obvious degradation of lignin.

It can be seen from figure that with the extension of hydrolysis time, CS all showed strong stretching band at 3400 cm^{-1} during hydrolysis. The O–H stretching band at 3400 cm^{-1} was weakened, which indicated that the hydrogen bond between cellulose molecules had been broken.

The absorption peak at 2925 cm^{-1} gradually decreased, which indicated that the aliphatic characteristics of decomposed CS decreased, and the extension of hydrolysis time accelerated the decomposition of carbohydrates and aliphatic compounds. The absorption peak at 1731 cm^{-1} disappeared after 8 h of hydrolysis, which illustrated that the ester bond formed by C = O of non-binding carbonyl carboxylate compounds in hemicellulose or lignin of CS disappeared. The absorption peaks at 1515 cm^{-1} and 1427 cm^{-1} were N–H deformation vibration of phthalimide and stretching vibration peak of lignin aromatic ring C = C, and the strength of the two peaks was significantly reduced through hydrolysis. This is mainly due to the removal of some lignin by further AH. With the advance of hydrolysis time, the absorption peak at 1374 cm^{-1}, 1245 cm^{-1}, and 1051 cm^{-1} decreased.

The absorption peak of the acetyl ester bond at 1245 cm^{-1} disappeared, mainly due to the decomposition of CS cellulose, which weakened its absorption intensity. The disappearance of the above peak after hydrolysis indicated that saponification destroyed part of the linkage between cellulose, hemicellulose, and lignin. It led to the release of part of cellulose and hemicellulose, the reduction of the steric hindrance between molecules, and increase of the porosity, which was beneficial for improving the fermentable performance of CS and promoting the biogas production. Zhang studied the effect of aerobic deterioration on methane production through AD of CS silage following 9 days air exposure of silage, and its effect of stover degradation was similar to the results of this study (Zhang et al., 2018). Therefore, AH can promote the decomposition of CS.

3.2. Effect of aerobic hydrolysis time on methane production

After charging and inoculation, all the reactors started smoothly and produced biogas normally. The AD experiment lasted for 26 d. At the end of the test, all treatments were close to stopping biogas production, and the pH after AD was in the range of 7.04–7.09. The cumulative methane production data in the test process were fitted using the first order model and modified Gompertz model for methane production. In the fitting process, the VS methane production rate (P_∞) was used as evaluation indicators (Abreu et al., 2019; Wainaina et al., 2020a).

The cumulative methane production and fitting curve of each treatment were shown in Fig. 3, and the fitting parameters of the equation were shown in Table 2. The R^2 ranges of the first order model and Gompertz model were 0.979–0.985 and 0.997–0.999, respectively, which indicated that both models can simulate the cumulative methane production well, and the Gompertz model was more suitable for it. With the progress of AH, the cumulative methane production first increased and then decreased, which reached the maximum value at 16 h of hydrolysis with 268.75 ml/g VS. The reason is that during the hydrolysis process, the accumulation of soluble organic matter did not change much in 16–24 h. But easy to be used insoluble solid organic matter was still converted into CO_2 and other substances through aerobic respiration during this period, which made the substrate used for methane production lose, thus reducing the methane production efficiency of the Z43-24. Moreover, Z43-16 had the maximum hydrolysis rate and constant, which indicated that it was appropriate to transform the AH process to AD at 16 h. In order to overcome the problem of low methane production rate of CS, cow manure and chicken manure, Wang indicated that mixing the materials could promote the hydrolysis process of AD and improve the start-up efficiency of the system (Fuchs et al., 2018; Wang et al., 2020b).

The cumulative methane production reached 202.10 ml/g VS, which was lower than that of this study. The main reason is that appropriate AH time can effectively degrade the components of lignocellulose, provide sufficient nutrition for the initial stage of AD, and increase the accessibility of hydrolases to cellulose in the fermentation system (Zhao et al., 2019). The analysis of variance for the methane yield was shown in supplementary file. The F value and p-value of between groups were 35.58 and 0.0001, respectively, which indicated that the interaction of hydrolysis time on methane yield was significant. Therefore, the methane production performance can be significantly improved after AH.

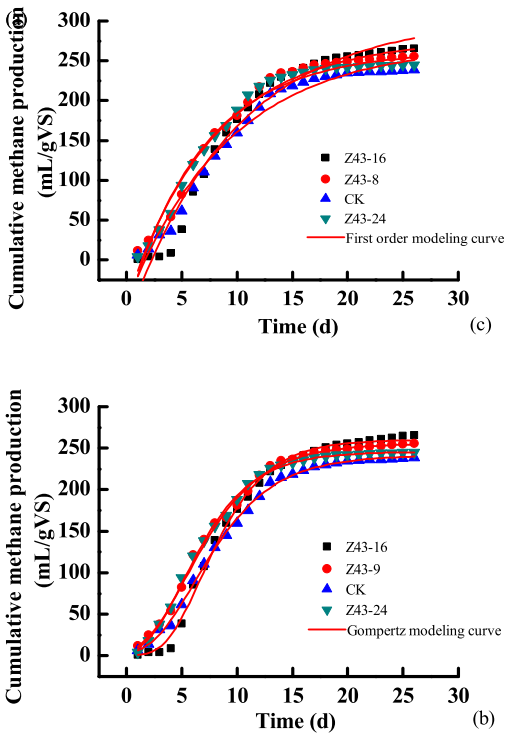


Fig. 3. Scatter and fitting curve of cumulative methane production; (a) First order modeling fitting of methane production with different aerobic hydrolysis time, and (b) Gompertz modeling fitting of methane production with different aerobic hydrolysis time.

3.3. Degradation of solid organic matter

Fig. 4 showed the degradation of organic solids and the content of methane at the end of AD. With the extension of hydrolysis time, the methane content increased first and then decreased, and the maximum value appeared at 16 h of hydrolysis, which reached 53.02%. The degradation rates of TS and VS of Z43-8, Z43-16 and Z43-24 increased gradually with the prolongation of hydrolysis time, which were all higher than that of CK. The degradation rates of TS and VS of Z43-16 were 60.23% and 60.58% respectively, which were 9.43% and 5.25% higher than those of CK. It indicated that the utilization rate of CS can be improved through AH, and the reason may be that the structure of lignin is destroyed during AH. The shielding effect of lignin on cellulose and hemicellulose partially disappeared, and the structure of lignocellulose of CS became loose, which was more easily used by methanogens extracellular enzymes. Aeration of the reaction system in AH stage not only increased the dissolved oxygen in the hydrolysate, but also promoted its heat and mass transfer performance. Moreover, the contact area between lignocellulose and methanogens was increased, which provided sufficient raw materials for the physiological activities of microorganisms and accelerated the growth and reproduction of methanogens (Wang et al., 2020a). Although the degradation rate of TS and VS of Z43-24 was the highest, it can be seen from the previous section that the methane production was the highest after 16 h of AH. Therefore, the increase of degradation rate of TS and VS did not necessarily improve the methane production of the system. AH of CS for 16 h can not only improve the degradation rate of TS and VS, but also avoid excessive degradation and unnecessary loss.

Table 2
Kinetic parameters from mathematically fitted profiles obtained for methane production at different treatment.

Treatment	Gompertz model				First-order model			
	P_∞(mL/g VS)	R_m(mL/d)	λ (d)	R^2	B_∞(mL/g VS)	K_H(d^{-1})	L_p(d)	R^2
CK	252.59	20.98 ± 0.41	1.46 ± 0.12	0.998	270.46 ± 9.21	0.10 ± 0.009	1.31 ± 0.22	0.984
Z43-8	265.52	24.78 ± 0.68	1.55 ± 0.14	0.997	276.71 ± 7.22	0.13 ± 0.010	1.38 ± 0.20	0.982
Z43-16	268.75	26.37 ± 0.56	1.69 ± 0.10	0.998	281.80 ± 5.44	0.14 ± 0.010	1.50 ± 0.16	0.985
Z43-24	253.42	22.52 ± 0.46	2.20 ± 0.11	0.999	271.45 ± 9.80	0.11 ± 0.010	1.67 ± 0.23	0.979

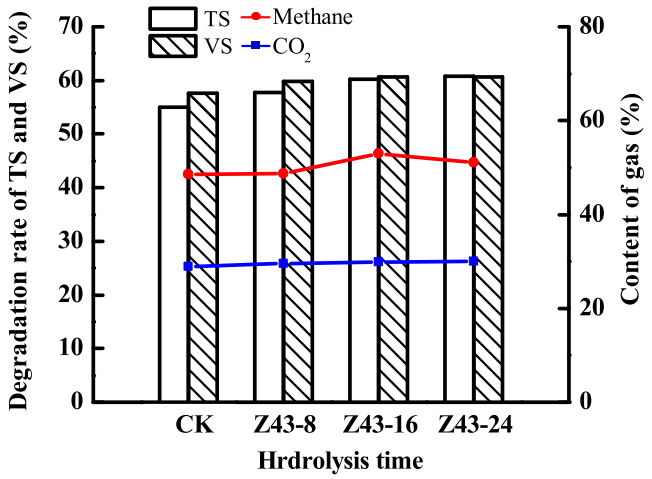


Fig. 4. Degradation of organic solids and the content of methane and CO_2 at the end of anaerobic digestion.

3.4. Energy consumption analysis

In this study, a 100 m^3 anaerobic reactor was taken as an example. To ensure the efficient operation of the reactor, the TS of fermentation substrate entering the system was 8%. Based on the Eq. (3) and Eq. (4), the heat consumption required for the increase of fermentation raw material temperature was shown in supplementary file. According to the original design parameters of the reactor, the heat consumption of each part of the reactor envelope can be calculated by the Eq. (7)- Eq. (11). The monthly heat consumption of AH and AD reactor was shown in supplementary file, respectively.

The calorific value of methane and CS is 890.31 kJ/mol and 17000 kJ/kg, respectively (Edwiges et al., 2018; Wainaina et al., 2019). In the storehouse, the annual temperature difference of raw materials is small, so the initial temperature of CS is assumed to be 15 °C. The difference between the total energy of methane output and the net energy consumption of input system (including raw material crushing, temperature increasing, and energy consumption in AH and AD reactor) is defined as the theoretical net productivity of the system. The ratio of theoretical productivity of each system to the initial total energy of CS is defined as energy conversion rate. Moreover, the ratio of theoretical net productivity of the system to total input net energy consumption is defined as embodied energy yield.

Fig. 5(a) showed the proportion of net energy consumption of AD after AH. With the extension of hydrolysis time, the proportion of crushing energy consumption fluctuated little, with the maximum of Z43-16 (12.93%) and the minimum of Z43-24 (8.47%). The hydrolysis energy consumption was the lowest, which was mainly related to the hydrolysis time and the monthly average temperature. The proportion of energy consumption of heating feed solution and AD was relatively large. In the proportion of heating, Z43-16 was the largest with 71% in July, and Z43-24 h was the smallest with 32.28% in January. However, the initial temperature of raw materials was closely related to the environment, so it was not easy to change this proportion.

The proportion of energy consumption of AD was closely related to the fermentation cycle. The maximum was 59.91% in January of Z43-8, and the minimum was 14.49% of Z43-16 in July, which mainly due to the shorter AD time reachingP_{90}. Elalami mentioned that biochemical methane potential (BMP) was not designed to scale up or estimate costs of AD system, the effects of any pretreatment were highly dependent on the AD process (Elalami et al., 2020). In different AD environment, the methanogenic ability of microorganisms is different. For example, compared with sequencing batch AD experiment of lignin-like substances, the hydraulic retention time of continuous AD is shorter, the degradation rate of lignocellulose is higher, and its AD system is more stable (Velez-Suberbie et al., 2018; Wainaina et al., 2020a). Therefore, there are some differences between the performance and net energy efficiency of methane production in laboratory scale and actual biogas engineering.

The energy conversion rate of AD after AH was shown in Fig. 5(b). From the perspective of hydrolysis time, all conversion rates of Z43-16 showed good performance, and the peak values of high-quality energy yield (445.41%), system energy conversion rate (17.91%), and CS energy conversion limit (18.66%) all appeared in July, which further indicated the hydrolysis effect was the best when AH was carried out to 16 h. When the temperature was warmed to a certain temperature, the raw materials can be better hydrolyzed and mix evenly (Anthraper et al., 2018; Khan et al., 2019). Therefore, the process of AH for 16 h can obtain better comprehensive properties.

4. Conclusions

Aerobic hydrolysis can effectively improve the degradation rate of lignocellulose. After 24 h of hydrolysis, lignin, cellulose, and hemicellulose of CS were degraded by 8.06%, 27.38%, and 35.73%, respectively. The methane production performance and maximum hydrolysis rate and constant were higher when the hydrolysis was carried out for 16 h. The proportion of energy consumption for crushing and hydrolysis was relatively low, while that for heating and AD was relatively high. In the biogas engineering, it is suggested that the hydrolysis temperature can be controlled at 43 °C and transferred to AD at 16 h of AH.

CRediT authorship contribution statement

Jingbo Qu: Conceptualization, Supervision, Writing - original draft, Funding acquisition, Project administration. **Yong Sun:** Conceptualization, Supervision, Writing - original draft, Funding acquisition, Project administration. **Mukesh Kumar Awasthi:** Supervision, Validation, Visualization. **Yuyingnan Liu:** Supervision, Conceptualization, Data curation. **Xinrui Xu:** Data curation, Formal analysis. **Xianghui Meng:** Supervision, Conceptualization, Data curation. **Hongqiong Zhang:** Conceptualization, Supervision, Writing - original draft, Funding acquisition, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

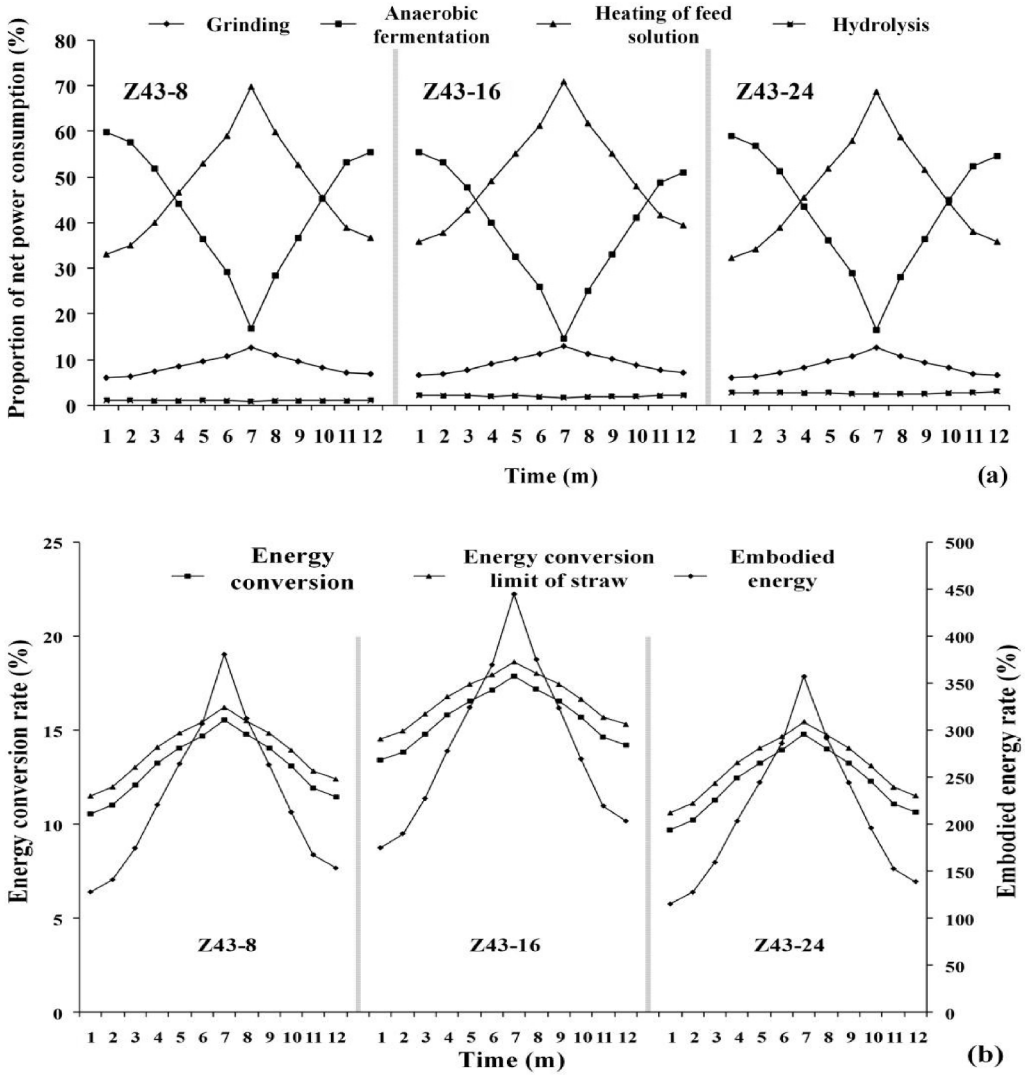


Fig. 5. Energy accounting of anaerobic digestion after aerobic hydrolysis; (a) net power consumption, and (b) energy conversion rate.

the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (52076034), National Key Research and Development Program Strategic Key Special Subject of International Science and Technology Innovation Cooperation (2018YFE0206600), National Key R&D Program of China (2019YFD1100603), and the Guangdong Provincial Key Laboratory of New and Renewable Energy Research and Development (E039kf0101).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2020.124332.

References

Abreu, A.A., Tavares, F., Alves, M.M., Cavaleiro, A.J., Pereira, M.A., 2019. Garden and food waste co-fermentation for biohydrogen and biomethane production in a two-step hyperthermophilic-mesophilic process. Bioresour. Technol. 278, 180–186.

Anthraper, D., McLaren, J., Baroutian, S., Munir, M.T., Young, B.R., 2018. Hydrothermal deconstruction of municipal solid waste for solid reduction and value production. J. Cleaner Prod. 201, 812–819.

Awasthi, M.K., Ravindran, B., Sarsaiya, S., Chen, H., Wainaina, S., Singh, E., Liu, T., Kumar, S., Pandey, A., Singh, L., Zhang, Z., 2020. Metagenomics for taxonomy profiling: tools and approaches. Bioengineered. 11, 356–374.

Bayrakdar, A., Tilahun, E., Calli, B., 2016. Biogas desulfurization using autotrophic denitrification process. Appl. Microbiol. Biotechnol. 100 (2), 939–948.

Bremond, U., de Buyer, R., Steyer, J.-P., Bernet, N., Carrere, H., 2018. Biological pretreatments of biomass for improving biogas production: an overview from lab scale to full-scale. Renew. Sust. Energ. Rev. 90, 583–604.

De la Rubia, M.A., Villamil, J.A., Rodrigues, J.J., Borja, R., Mohedano, A.F., 2018. Mesophilic anaerobic co-digestion of the organic fraction of municipal solid waste with the liquid fraction from hydrothermal carbonization of sewage sludge. Waste Manage. 76, 315–322.

Du, H., Li, F., 2017. Characteristics of dissolved organic matter formed in aerobic and anaerobic digestion of excess activated sludge. Chemosphere 168, 1022–1031.

Duan, N., Zhang, D.J., Lin, C., Zhang, Y.F., Zhao, L.Y., Liu, H.B., Liu, Z.D., 2019. Effect of organic loading rate on anaerobic digestion of pig manure: Methane production, mass flow, reactor scale and heating scenarios. J. Environ. Manage. 231, 646–652.

Edwiges, T., Frare, L., Mayer, B., Lins, L., Triolo, J.M., Flotats, X., Costa, M.S.S.D., 2018. Influence of chemical composition on biochemical methane potential of fruit and vegetable waste. Waste Manage. 71, 618–625.

Elalami, D., Monlau, F., Carrere, H., Abdelouahdi, K., Charbonnel, C., Oukarroum, A., Zeroual, Y., Barakat, A., 2020. Evaluation of agronomic properties of digestate from macroalgal residues anaerobic digestion: Impact of pretreatment and co-digestion with waste activated sludge. Waste Manage. 108, 127–136.

Fuchs, W., Wang, X., Gabauer, W., Ortner, M., Li, Z., 2018. Tackling ammonia inhibition for efficient biogas production from chicken manure: Status and technical trends in Europe and China. Renew. Sust. Energ. Rev. 97, 186–199.

Kavitha, S., Brindha, G.M.J., Gloriana, A.S., Rajashankar, K., Yeom, I.T., Banu, J.R., 2016. Enhancement of aerobic biodegradability potential of municipal waste activated sludge by ultrasonic aided bacterial disintegration. Bioresour. Technol. 200, 161–169.

Khan, T.A., Saud, A.S., Jamari, S.S., Ab Rahim, M.H., Park, J.W., Kim, H.J., 2019. Hydrothermal carbonization of lignocellulosic biomass for carbon rich material preparation: A review. Biomass Bioenerg. 130, 105385.
Li, L., Pagilla, K.R., 2017. Biomass density-function relationships in suspended growth biological processes - A critical review. Water Res. 111, 274–287.
Li, Y., Chen, Y.G., Wu, J., 2019. Enhancement of methane production in anaerobic digestion process: A review. Appl. Energ. 240, 120–137.
Li, Y., Zhang, Z., Zhu, S., Zhang, H., Zhang, Y., Zhang, T., Zhang, Q., 2018. Comparison of bio-hydrogen production yield capacity between asynchronous and simultaneous saccharification and fermentation processes from agricultural residue by mixed anaerobic cultures. Bioresour. Technol. 247, 1210–1214.
Lim, J.W., Wang, J.-Y., 2013. Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste. Waste Manage. 33 (4), 813–819.
Long, X., Wang, H., Wang, C., Li, X., 2019. The synergistic effect of biophoto anode for the enhancement of current generation and degradation. Environ. Technol. 1–11.
Ma, Y.W., Li, W.Z., Luo, L.N., Zhang, H.Q., Li, P.F., Zhang, X., Xu, X., Zhang, B., 2017. Effects of aerobic pre-treatment on anaerobic batch digestion of corn straw. J. Biobased Mater. Bioenergy. 11 (6), 622–628.
Miner, G., 2006. Standard methods for the examination of water and wastewater, 21st Edition. J. Am. Water Work Assoc. (1), 130.
Park, J.-G., Shin, W.-B., Shi, W.-Q., Jun, H.-B., 2019. Changes of bacterial communities in an anaerobic digestion and a bio-electrochemical anaerobic digestion reactors according to organic load. Energies 12 (15).
Peng, W., Lu, F., Hao, L.P., Zhang, H., Shao, L.M., He, P.J., 2020. Digestate management for high-solid anaerobic digestion of organic wastes: A review. Bioresour. Technol. 297.
Schiltz, C.J., Adams, M.C., Chappie, J.S., 2020. The full-length structure of *Thermus scotoductus* OLD defines the ATP hydrolysis properties and catalytic mechanism of Class 1 OLD family nucleases. Nucleic Acids Res. 48 (5), 2762–2776.
Shi, X., Zuo, J., Li, B., Yu, H., 2020. Two-stage anaerobic digestion of food waste coupled with in situ ammonia recovery using gas membrane absorption: Performance and microbial community. Bioresour. Technol. 297, 122458.
Srisowmeya, G., Chakravarthy, M., Devi, G.N., 2020. Critical considerations in two-stage anaerobic digestion of food waste - A review. Renew. Sust. Energ. Rev. 119.
Sun, Y., Qu, J., Li, R., Li, W., Wang, Z., Chu, X., 2018. Optimization of the enzyme production conditions of *Bacillus licheniformis* and Its effect on the degradation of corn straw. J. Biobased Mater. Bioenergy. 12 (5), 432–440.
Sun, Y., Wang, Z., Liu, Y., Meng, X., Qu, J., Liu, C., Qu, B., 2020a. A review on the transformation of furfural residue for value-added products. Energies. 13, 21.
Sun, Y., Zhang, Z., Sun, Y., Yang, G., 2020b. One-pot pyrolysis route to Fe−N-Doped carbon nanosheets with outstanding electrochemical performance as cathode materials for microbial fuel cell. Int. J. Agric. Biol. Eng.
Van Soest, P.J., 1967. Use of detergents in the analysis of fibrous feed. IV. Determination of cell-wall constituents. J. Assoc. Off. Anal. Chem. 50, 50–55.
Velez-Suberbie, M.L., Betts, J.P.J., Walker, K.L., Robinson, C., Zoro, B., Keshavarz-Moore, E., 2018. High throughput automated microbial bioreactor system used for clone selection and rapid scale-down process optimization. Biotechnol. Prog. 34 (1), 58–68.
Wang, C., Zhang, J., Hu, F., Zhang, S., Lu, J., Liu, S., 2020a. Bio-pretreatment promote hydrolysis and acidification of oilseed rape straw: Roles of fermentation broth and micro-oxygen. Bioresour. Technol. 308, 123272.
Wang, X., Yan, R., Zhao, Y., Cheng, S., Han, Y., Yang, S., Cai, D., Mang, H.-P., Li, Z., 2020b. Biogas standard system in China. Renew. Energy 157, 1265–1273.
Wang, Z., Cheng, Q., Liu, Z., Qu, J., Chu, X., Li, N., Noor, R.S., Liu, C., Qu, B., Sun, Y., 2019. Evaluation of methane production and energy conversion from corn stalk using furfural wastewater pretreatment for whole slurry anaerobic co-digestion. Bioresour. Technol. 293, 121962.
Wang, Z.T., Gong, Z.Q., Wang, W., Zhang, Z., 2020c. Study on combustion characteristics and the migration of heavy metals during the co-combustion of oil sludge char and microalgae residue. Renew. Energy. 151, 648–658.
Wainaina, S., Lukitawesa, Awasthi, M.K., Taherzadeh, M.J., 2019. Bioengineering of anaerobic digestion for volatile fatty acids, hydrogen or methane production: a critical review. Bioengineered 10, 437–458.
Wainaina, S., Awasthi, M., Sarsaiya, S., Chen, H., Singh, E., Kumar, A., Ravindran, B., Awasthi, S., Liu, T., Duan, Y., Kumar, S., Zhang, Z., Taherzadeh, M., 2020a. Resource recovery and circular economy from organic solid waste using aerobic and anaerobic digestion technologies. Bioresour. Technol. 301, 122778.
Wainaina, S., Awasthi, M.K., Horvath, I., Taherzadeh, M., 2020b. Anaerobic digestion of food waste to volatile fatty acids and hydrogen at high organic loading rates in immersed membrane bioreactors. Renew. Energ. 152, 1140–1148.
Wu, C., Yu, M., Huang, Q., Ma, H., Gao, M., Wang, Q., Sakai, K., 2018. Stimulation of methane yield rate from food waste by aerobic pre-treatment. Bioresour. Technol. 261, 279–287.
Yang, F., Bai, L., Li, P., Li, Q., Luo, L., Li, W., 2019a. Improved methane production and sulfate removal by anaerobic co-digestion corn stalk and levulinic acid wastewater pretreated by calcium hydroxide. Sci. Total Environ. 691, 499–505.
Yang, Q., Wang, D., Huo, D., Lin, L., Liu, Q., Liu, Z., Zhan, F., 2019b. Structural characterization of the bagasse lignin pretreated using solid alkali. Nord. Pulp Paper Res. J. 34 (4), 558–566.
Yu, Q., Liu, R., Li, K., Ma, R., 2019. A review of crop straw pretreatment methods for biogas production by anaerobic digestion in China. Renew. Sust. Energ. Rev. 107, 51–58.
Zhang, H., Wu, J.W., Gao, L.J., Yu, J.D., Yuan, X.F., Zhu, W.B., Wang, X.F., Cui, Z.J., 2018. Aerobic deterioration of corn stalk silage and its effect on methane production and microbial community dynamics in anaerobic digestion. Bioresour. Technol. 250, 828–837.
Zhang, T., Jiang, D.P., Zhang, H., Lee, D.J., Zhang, Z.P., Zhang, Q.G., Jing, Y.Y., Zhang, Y., Xia, C.X., 2020. Effects of different pretreatment methods on the structural characteristics, enzymatic saccharification and photo-fermentative bio-hydrogen production performance of corn straw. Bioresour. Technol. 304, 122999.
Zhao, M., Shi, D., Lu, X., Zong, H., Zhuge, B., Ji, H., 2019. Ethanol fermentation from non-detoxified lignocellulose hydrolysate by a multi-stress tolerant yeast *Candida glycerinogenes* mutant. Bioresour. Technol. 273, 634–640.
Zou, J.T., Pan, J.Y., He, H.T., Wu, S.Y., Xiao, N.D., Ni, Y.J., Li, J., 2018. Nitrifying aerobic granular sludge fermentation for releases of carbon source and phosphorus: The role of fermentation pH. Bioresour. Technol. 260, 30–37.